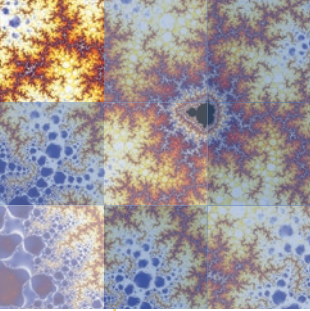




Operations with Complex Numbers

Complex numbers can be added, subtracted, multiplied, and divided.

Adding and Subtracting Complex Numbers

The commutative and associative properties of addition and multiplication, as well as the distributive property, hold for complex numbers.

Adding and Subtracting Complex Numbers

Let a , b , c , and d be real numbers.

Addition To add complex numbers, add the real parts and then add the imaginary parts.

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

Subtraction To subtract complex numbers, subtract the real parts and then subtract the imaginary parts.

$$(a + bi) - (c + di) = (a - c) + (b - d)i$$

Closure Properties for the Complex Numbers

The set of complex numbers is closed under addition, subtraction, multiplication, and division.

EXAMPLE 1

Add or subtract, and then simplify.

A $(3 + 7i) + (-9 - 4i)$

SOLUTION

$$\begin{aligned}(3 + 7i) + (-9 - 4i) &= (3 + (-9)) + (7 + (-4))i \\ &= -6 + 3i\end{aligned}$$

B $(-5 - 2i) - (8 + 11i)$

SOLUTION

$$\begin{aligned}(-5 - 2i) - (8 + 11i) &= (-5 - 8) + (-2 - 11)i \\ &= -13 - 13i \quad \blacksquare\end{aligned}$$

Multiplying Complex Numbers

Use the distributive property to multiply complex numbers.

► **Remember** $i^2 = (\sqrt{-1})^2 = -1$

EXAMPLE 2

Multiply.

$$7i(3 + 2i)$$

SOLUTION

Use the distributive property.

$$\begin{aligned}7i(3 + 2i) &= 7i \cdot 3 + 7i \cdot 2i \\ &= 21i + 14i^2 \\ &= 21i + 14(-1) \\ &= 21i - 14 \\ &= -14 + 21i \quad \blacksquare\end{aligned}$$